Traumatic Brain Injury (TBI)

Guideline Title: Traumatic Brain Injury (TBI)	Guideline Number: GUID-3203-T014

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 6/8/2021	Developed by: Air Care Team
Approved by: Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature:	Signature:
Department Numbers: 3203	

CXXXII. **PURPOSE:** To minimize increases in intracranial pressure and to maintain optimal cerebral oxygenation and tissue perfusion.

PATHOPHYSIOLOGY:

Brain injury as a result of head trauma occurs by both of the following: Primary "impact" damage as the immediate consequence of the injury; and Secondary complications of impact damage, such as hematoma (epidural, SAH, or intracerebral) or cerebral swelling, sometimes with herniation syndromes. Blunt head trauma usually occurs through one of two mechanisms: concussive-compressive and acceleration-deceleration injuries. Concussive-compressive injuries typically result when a blunt object strikes the head, and localized type of injury is the usual outcome. Acceleration-deceleration is commonly a result of motor vehicle collisions. In a traffic crash the victim's head rapidly stops but the brain continues to travel forward, impacting on the inner skull. Widespread, diffuse axonal injuries may result secondary to shearing forces. Concussion, a transient loss of consciousness, without obvious anatomic injury, is a less severe form of diffuse injury.

GCS is a reliable indicator of brain injury in the field. Minor TBI: GCS of 13-15. Moderate TBI: GCS of 9-12. Severe TBI: GCS of 3-8. Indicators of impending herniation include: unconscious & unresponsive with extensor posturing, asymmetric, dilated, or nonreactive pupils, or progressive neurologic deterioration (decrease in the GCS score of more than two points from the patient's prior best score in patients with an initial GCS of < 9).

CXXXIII. **DEFINITIONS:**

A. GCS – Glasgow Coma Scale
B. ICP – Intracranial pressure
C. SBP – Systolic Blood Pressure
D. TBI – Traumatic Brain Injury

CXXXIV. **DEPARTMENT GUIDELINE:**

A. Assess and manage airway, breathing, and circulation. Determine GCS and assess vital signs.
B. If pt has a GCS of < 9 and/or has a deteriorating LOC secure an advanced airway *per GUID-G008 Rapid Sequence Induction for Endotracheal Intubation and GUID-PR020 Endotracheal Intubation.*
C. Maintain full spinal precautions.

D. Assess for:
    i. Blood loss, soft tissue injury, depression of skull indicating fracture, periorbital and/or mastoid ecchymosis, presence of otorrhea and/or rhinorrhea, and seizure activity.
    ii. Note movement of extremities (normal, abnormal, or absent).
    iii. Presence of Cushing's reflex: hypertension, bradycardia, and an irregular respiratory pattern, which may indicate rising ICP.
    iv. **Signs of herniation:** if no hypoxia, hypotension, over-sedation, or other cause, signs of herniation include unilateral dilated pupil, nonreactive pupil, decrease in GCS of 2 or more in a patient with a GCS less than 9, and/or spontaneous posturing, CT scan showing herniation or significant midline shift.

E. For patients with evidence of herniation:
    i. <u>For interfacility transfers only and proven by CT scan with intracranial hemorrhage</u> (subdural hematoma, epidural hematoma, intraparenchymal bleed, etc.)
        1. Assess for other causes of hypoxia, hypotension, or over-sedation.
        2. Administer **Hypertonic 3% Saline**
            a. **Adult: 500ml IV** over 10 minutes
            b. **Pediatric: 3-4ml/kg IV** over 10 minutes
        3. If sending facility prefers, consider obtaining and administering **Mannitol 1 gram/kg IV** over 5-10 minutes using a filter on IV tubing if adequate blood pressures.
        4. Controlled hyperventilation can be considered to decrease ETCO2 30-35 mmHg.

F. If mechanically ventilated, maintain ETCO2 35-40 mmHg and avoid PEEP >5 unless needed for adequate oxygenation as it may contribute to increased ICP

G. Elevate head of the bed 30 degrees unless contraindicated.

H. Obtain IV/IO access. Treat hypotension with fluids per *GUID-M011 – Hypotension.* Maintain SBP >90 mmHg for optimal cerebral perfusion.

I. Consider gastric decompression per *GUID-PR010 - Gastric Tube Insertion.*

J. If patient taking anti-coagulants, consider reversal agents after discussion from sending facility.
    i. Administer appropriately ordered reversal agent for patients on anticoagulant medications per transferring or receiving facility orders. (For example: anti-inhibitor coagulant complex [FEIBA], idarucizumab [Praxbind], Vitamin K, coagulation factor Xa recombinant [Andexanet alpha]).
    ii. For patients taking Warfarin & INR>1.4, consider discussing with sending physician
        1. **Vitamin K 1-2.5mg PO or 10mg IVPB** of no prosthetic heart valve
        2. **FEIBA 1000 units/20ml IVPB** over 15 minutes
        3. Fresh frozen plasma 2 units in non-surgical observation only patients unless patient requires fluid restriction.
    iii. For patients taking oral factor Xa Inhibitors (Apixaban/Eliquis, Edoxaban/Sayvaysa, Rivaroxaban/Xarelto), consider discussing with sending physician
        1. **FEIBA 2000 units/40ml IVPB** over 15 minutes
        2. **Tranexamic acid 1-gram IVPB** over 10 minutes
    iv. If FEIBA was ordered at sending facility:

1. Wait for the FEIBA medication before transferring the patient, even if it is a >20-minute delay. If the patient is at the point of herniating, call the receiving physician to discuss if they would prefer you to transfer without delay.
2. The IV access you use for FEIBA is dedicated to only FEIBA for time of administration.
3. Ensure the MD/RN at receiving facility is aware of the time that FEIBA was given.
4. Observe for infusion reactions, thromboembolic events, & allergic reactions.

v. For patients taking Dabigatran/Pradaxa, consider discussing with sending physician

1. **Idarucizumab 2.5gram IVPB** every 10 minutes x 2 doses

K. Treat pain and/or anxiety per *GUID-G005 - Analgesia and Sedation Management*.

L. For combative patients secondary to head trauma

i. Ensure hypoxia and hypotension are addressed.

ii. Apply physical restraints if needed to ensure safety per *GUID-PR021 – Restraints.*

iii. Additional considerations per *GUID-G006 – Anxiety/Combative Patient Management.*

1. Avoid benzodiazepines and utilize ketamine or haloperidol in this patient population

M. Consider premedicating patient per *GUID-G004 - Nausea/Vomiting/Motion Sickness.*

N. For seizures, treat per *GUID-M015 – Seizures*.

O. Consider prophylactic anticonvulsants for confirmed intracranial hemorrhage, in the immediate post-hemorrhagic phase with **Keppra 1 gram IV** over 2-5 minutes

CXXXV. **DOCUMENTATION:**

A. EMS Charts

CXXXVI. **REFERENCES:**

A. Badjatia, N., Carney, N., Crocco, TJ, et al. (2008). Guidelines for prehospital management of traumatic brain injury 2nd edition. *Prehosp Emerg Care.* 12 Suppl 1:S1.

B. Davis, DP., Peay, J., Serrano, JA., et al. (2005). The impact of aeromedical response to patients with moderate to severe traumatic brain injury. *Ann Emerg Med.* 46, 115.